DP

DapurPlan

SDC HACKATHON 2026

Cook to demand. Not guesswork.

A daily decision workspace for food businesses: forecast servings, calculate ingredients, review the plan.

TEAM S.E.E.S • DOMAIN 2 • SDG 12

THE DECISION WE HELP MAKE

What should we prepare for tomorrow?

DapurPlan connects recent sales with recipes and usable stock—so a person can make a better-informed preparation decision.

FORECAST
servings

CALCULATE
ingredient needs

REVIEW
draft or approve

Honest starting point: today's model is trained and evaluated on synthetic data.

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Team S.E.E.S | UMT × UiTM | 3 members

A daily decision is fragmented across records

Small kitchens often have the information—but not in one place or one decision flow.

THE OPERATOR'S MORNING

01 **Recall yesterday**

What sold? What was not entered?

02 **Estimate demand**

Will tomorrow be like yesterday or last week?

03 **Translate to ingredients**

How much rice, chicken, eggs, oil?

04 **Check what is usable**

What remains, and what expires first?

THE FRICTION

Manual hand-offs make the plan hard to calculate, explain, and update when sales or stock changes.

WHAT WE ARE NOT ASSUMING

We do not claim a measured waste rate, stockout rate, or market size. These are problems to establish with real operators and baseline data.

One workflow from sales history to reviewed preparation

DapurPlan brings the daily records and decisions into a single workspace.

1. RECORD

Sales • recipes • prices • stock

Capture daily servings, recipes and ingredient quantities, purchase costs, and dated stock batches.

2. FORECAST

One day ahead

Estimate next-day servings for five menu items from a complete seven-day history.

3. TRANSLATE

Menu servings → ingredients

Scale each recipe to planned servings and aggregate ingredient requirements.

4. RECONCILE

Need • available • shortage

Compare requirements with positive, usable inventory and show purchase pack guidance.

5. REVIEW

Draft → approved

Let the operator adjust quantities and confirm stock checks before owner approval.

THE RESULT: a traceable preparation decision that a person can review—not an automatic order or stock deduction.

What the operator sees—and decides

A practical demo path: enter history, inspect demand, adjust the plan, then approve.

TOMORROW'S PLAN / WORKSPACE FLOW

Forecast history complete • model ready — 5 MENU ITEMS

MENU	FORECAST	PREP	NEED / STOCK	STATUS
Chicken rice	36.4	37	Rice · covered	READY
Kuey teow	29.1	30	Egg · check	REVIEW
Nasi lemak	42.8	43	Coconut · short	BUY
Kopi O ais	54.6	55	Coffee · covered	READY
Iced lemon tea	39.2	40	Tea · covered	READY

SCHEMATIC EXAMPLE — values shown are illustrative, not live customer data.

The decision stays with the operator

A — Check the basis

Seven prior complete days; missing entries are shown, not silently zero-filled.

B — Adjust preparation

Forecast defaults are rounded up; the operator may change whole-serving quantities.

C — Review stock

Inspect usable batches, expiry warnings and ingredient shortages.

D — Approve deliberately

Owner confirms stock review; approval does not place orders or deduct stock.

A transparent one-day-ahead forecast

Five menu items • seven complete prior days • simple, reviewable sales features.

FEATURES CREATED FOR EACH MENU ITEM

DAY OF WEEK	YESTERDAY	LAST WEEK	7-DAY MEAN
Monday = 0 Sunday = 6	Most recent completed day	Same weekday 7 days back	Average of the previous 7 days

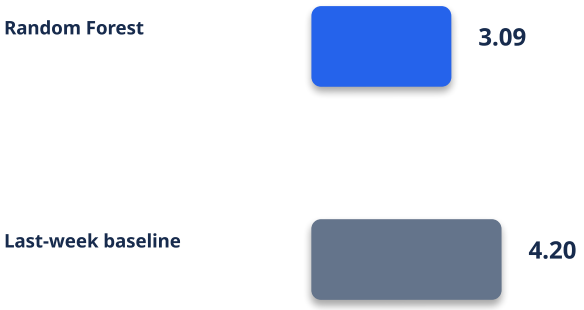
RANDOM FOREST • 300 TREES • MAX DEPTH 6

One-hot encodes menu identity and passes the numeric history features to a fitted pipeline.

No complete 7-day history? Forecast is blocked; missing sales are not treated as zero.

Synthetic holdout check

Mean absolute error • servings



3.09 vs 4.20 MAE on synthetic data only. Not a real-world performance claim.

Predictions are clipped at zero; preparation defaults round up to whole servings.

Translate a menu decision into ingredient quantities

The calculation is recipe-led, unit-aware, and explicit about what stock counts as available.

1 / SCALE EACH RECIPE LINE

ingredient need
= recipe quantity × planned servings ÷ recipe batch servings

2 / AGGREGATE SHARED INGREDIENTS

Rice for chicken rice + rice for nasi lemak = total rice requirement.

3 / COMPARE WITH USABLE STOCK

shortage = max(0, required − available)

WHAT COUNTS AS USABLE?

COUNTED	Positive batches with no expiry date, or expiry on/after preparation date.
EXCLUDED	Batches expired before preparation date.
FLAGGED	Unknown expiry and older count dates receive review warnings.
ROUNDED	Count ingredients are rounded up to whole pieces.
SUGGESTED	Shortage can be expressed in configured purchase pack sizes.

Approval records review; it does not deduct inventory or place an order.

The forecast is one part of a connected operating record

The surrounding data makes the recommendation explainable and keeps a trace of decisions.

SALES

Five menus • daily servings
Blank = not recorded
Zero = confirmed zero

RECIPES & COSTS

Ingredient lines • batch yield
Selling price • pack cost
Saved cost snapshots

INVENTORY

Dated ingredient batches
Count • expiry • pack size
Change history

EXPENSES

Daily category • amount
Optional note
Estimated net result

ROLE-AWARE REVIEW

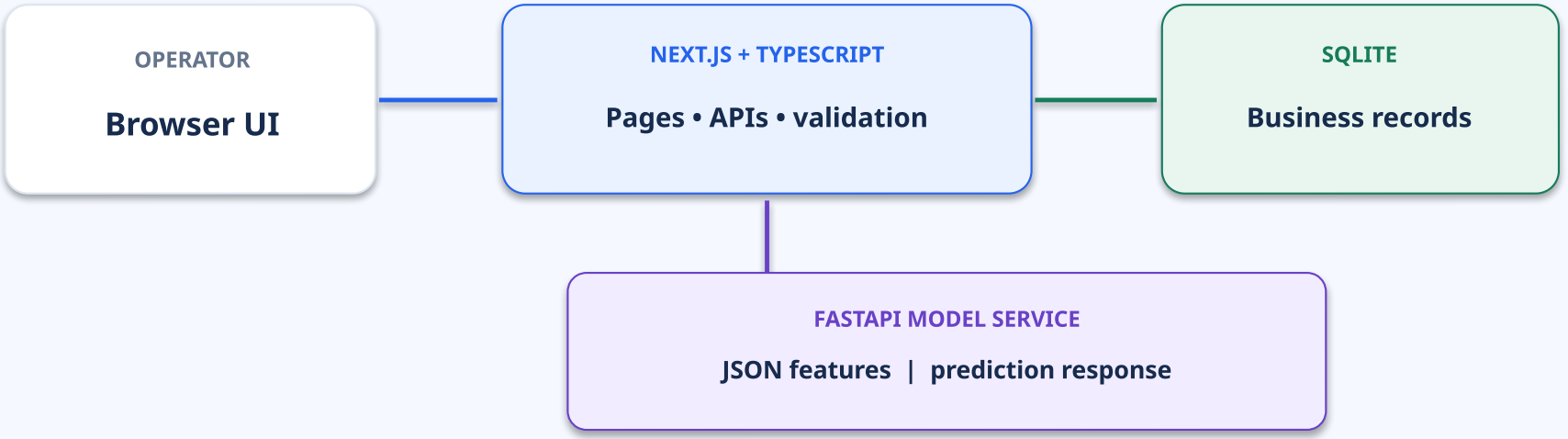
Owners manage recipes, prices and approval. Staff can record day-to-day sales and stock within role rules.

CSV WHEN A WORKBOOK IS EASIER

Supported data can be exported and imported with row-level validation. Planning is generated from source records and remains export-only.

A lightweight application with a deliberately separate model service

The web application owns business data; the prediction API receives prepared features only.



BUSINESS DATA BOUNDARY

SQLite stores recipes, sales, stock, plans and history. The forecast API has no database access.

DEPLOYMENT STATUS

Local setup is documented. RunPod CPU Pod deployment is optional; no live endpoint or Replit deployment is configured.

SDG 12 alignment is a hypothesis to measure—not a result to claim

The product is designed to support more deliberate preparation and purchasing decisions.

SDG 12

Responsible Consumption

A better-informed prep plan may help operators align production with demand and stock.

Potential contribution; not yet measured.

PILOT MEASUREMENT PLAN

Forecast error	MAE by menu; compare with last-week baseline.
Preparation variance	Planned servings vs. servings sold and remaining.
Ingredient outcome	Shortage flags, extra purchases and unused stock.
Workflow adoption	Days with complete records; time to create a plan.

Pilot baseline first; collect outcomes only with operator consent.

What is working today—and what still needs proof

Credibility comes from making the boundaries visible and validating the next step.

IMPLEMENTED

- Input checks for dates, quantities, units and required menu rows.
- Seven-day completeness gate; missing history is surfaced.
- Chronological model validation and a last-week baseline.
- Re-training excludes demo-provenance, incomplete and sold-out observations.
- Current code requires 84 complete non-demo days before retraining.

LIMITS TO ADDRESS

- The shipped model was trained and tested on synthetic data only.
- No real-restaurant accuracy or waste reduction is established.
- Fixed owner/staff demo accounts are not production authentication.
- SQLite is single-machine prototype storage; deployment persistence is unverified.
- RunPod and Replit are not currently live deployments.

The next milestone is a measured, operator-led pilot

Move from a working prototype to evidence in deliberate stages.

NOW

Prototype

- Daily records
- Forecast flow
- Recipe-scaled needs
- Review & approval

NEXT

Pilot readiness

- Secure accounts
- Persistent deployment
- Operator onboarding
- Consent & baseline

PILOT

Measure in context

- Track real sales
- Compare baseline
- Review stock outcomes
- Collect user feedback

LEARN

Improve safely

- Retrain only with sufficient data
- Monitor per-menu error
- Iterate with operators

Success means better decisions demonstrated with real operational data—not a model score alone.

TEAM S.E.E.S • DOMAIN 2 • SDG 12

Cook to demand.
Not guesswork.

We’re building a practical bridge between what sold, what a recipe needs, and what the kitchen already has.

Our ask: help us validate the workflow with food operators and real, consented sales data.

Universiti Malaysia Terengganu × Universiti Teknologi MARA • 3 members

github.com/Etticx/DapurPlan